ADITYA KUMAR TIWARY

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Professional Summary

AI-focused Software Engineering undergraduate with hands-on experience in **Machine Learning, Deep Learning, NLP, Generative AI, and LLM-based applications**. Skilled in building scalable backend systems, **cloud-native microservices**, and deploying end-to-end AI solutions using **LangChain**, **RESTful APIs**, and **containerization**.

Technical Skills

- **Languages & Databases:** Python, C++, JavaScript, SQL, MongoDB, PostgreSQL, Redis
- **Frameworks & Libraries:** Node.js, React.js, Express.js, Django, FastAPI, TensorFlow, Keras, PyTorch, scikit-learn, Pandas, NumPy, Hugging Face Transformers, LangChain
- **Cloud & DevOps:** Docker, Kubernetes, AWS (EC2, S3, Lambda), CI/CD (GitHub Actions), Linux
- **Core CS & AI:** Data Structures, Algorithms, OOP, Operating Systems, Computer Networking, Computer Vision, NLP, LLMs, RAG, Distributed Systems

Professional Experience

AI & Cloud Intern — IBM *May 2024 – August 2024*
Bangalore, India

- Developed and optimized a **real-time traffic monitoring system** using TensorFlow (computer vision), improving detection accuracy by **25%**.
- Reduced API response latency by **30%** through microservices optimization and inference pipeline tuning with Flask and Docker.
- Built a cloud-integrated dashboard handling **1000+ daily events** for monitoring logs and system performance on AWS (EC2, S3).
- Deployed containerized services ensuring **high availability and scalability** with CI/CD pipelines.

Projects

Smart News Analyzer (Distributed System) — GitHub *Jan 2026 – Feb 2026*
FastAPI, Django, Redis, Celery, React, NLP (spaCy, Hugging Face)

- Architected a **microservices-based architecture** for scraping and processing news from multiple sources.
- Implemented asynchronous processing using **Celery + Redis**, reducing data processing time by **40%**.
- Built NLP pipelines for **transformer-based summarization and sentiment analysis** to generate actionable insights.

AI Resume Analyzer (ATS Optimization Engine) — GitHub *Sept 2025 – Oct 2025*
Python, Django, React, NLP (BERT, TF-IDF)

- Created an AI-based system to parse and analyze resumes using **BERT embeddings and keyword extraction**.
- Developed a custom scoring algorithm improving keyword-job match accuracy by **40%**.
- Designed backend APIs for handling **concurrent requests and real-time processing** with Django REST Framework.
- Implemented dynamic feedback system to optimize resumes based on job descriptions.

Lung Cancer Detection System (Deep Learning) — GitHub *Jun 2025 – July 2025*
TensorFlow, Flask, Docker, OpenCV

- Developed a **CNN-based model with transfer learning** for detecting lung cancer from CT scan images.
- Applied preprocessing and augmentation pipelines to improve model performance and generalization.
- Containerized model using Docker and deployed as a **scalable REST API** with Flask on AWS EC2.

Education

B.E. Electronics & Communication Engineering 2022 – 2026
Sir M. Visvesvaraya Institute of Technology, Bangalore

Certifications & Achievements

- Solved **300+ DSA problems** on LeetCode (Rating: 1616)
- **3rd Place** – CertifyO Hackathon (AI-powered solution)
- Full Stack Development (MERN) – Udemy
- Gate C.S.E 2026 Qualified